

MATH 637: Mathematical Techniques in Data Science

Questionnaire

ABOUT YOU

Name: _____

☐ Undergrad in year: _____ ☐ MS student ☐ PhD student in year: _____ ☐ Other: _____

☐ Math/Stats ☐ Computer Science ☐ Engineering ☐ Other dept.: _____

GENERAL BACKGROUND

What is your level of comfort with the following topics?

Topic / Comfort level	Low	Medium	High
Linear algebra (e.g., matrix algebra, eigenvalues/eigenvectors, SVD)	☐	☐	☐
Probability (e.g., random variables, distributions, conditioning)	☐	☐	☐
Multivariable calculus (e.g., partial derivatives, gradients, chain rule)	☐	☐	☐
Programming in <i>some</i> language	☐	☐	☐
Python packages (e.g., NumPy, pandas, scikit-learn)	☐	☐	☐

DATA SCIENCE BACKGROUND

Prior course in statistics, data science, or machine learning? ☐ Yes ☐ No

If yes, which course(s)? _____

Practical data science experience (course, research, internship, or job)? ☐ Yes ☐ No

If yes, briefly describe: _____

How familiar are you with the following types of techniques?

Data science technique / Familiarity	Never heard of it	Heard of it	Used it
Dimension reduction (e.g., principal component analysis)	☐	☐	☐
Clustering (e.g., k-means algorithm)	☐	☐	☐
Regression (e.g., ordinary least squares, ridge regression)	☐	☐	☐
Classification (e.g., logistic regression, decision trees)	☐	☐	☐
Neural networks (e.g., multilayer perceptron)	☐	☐	☐
Model assessment (e.g., cross-validation, calibration)	☐	☐	☐

GOALS & INTERESTS

Do you aspire to work as a professional data scientist? ☐ Yes ☐ No ☐ Maybe

Interested in data science-related research (e.g., high-dimensional probability, optimization)? ☐ Yes ☐ No ☐ Maybe

One thing you'd like to get out of this course: _____

[Optional] A movie/book/album recommendation: _____